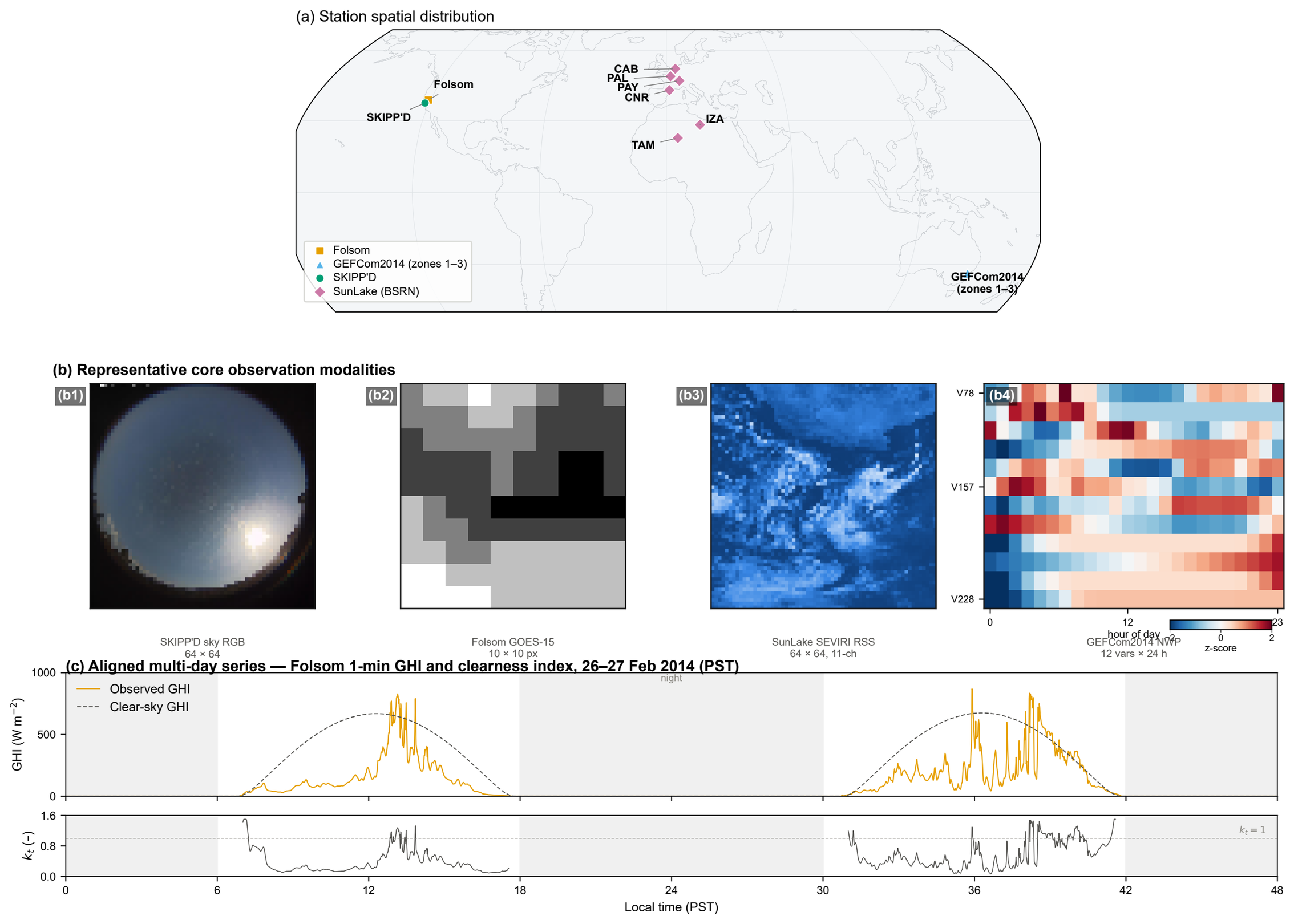

(a) Station spatial distribution
Folsom
SKIPP'D
CAB
PAL
PAY
CNR
IZA
TAM
Folsom
GEFCom2014 (zones 1–3)
SKIPP'D
SunLake (BSRN)
GEFCom2014
(zones 1–3)
(b) Representative core observation modalities
(b1)
(b2)
(b3)
(b4)
V78
V157
V228
0
12
23
hour of day
−2
0
2
z-score
SKIPP'D sky RGB
64 × 64
Folsom GOES-15
10 × 10 px
SunLake SEVIRI RSS
64 × 64, 11-ch
GEFCom2014 NWP
12 vars × 24 h
(c) Aligned multi-day series — Folsom 1-min GHI and clearness index, 26–27 Feb 2014 (PST)
night
Observed GHI
Clear-sky GHI
GHI (W m−2)
1000
500
0
1.6
0.8
0.0
kt (–)
kt = 1
0
6
12
18
24
30
36
42
48
Local time (PST)